CREATIVE VIDEO TIPS · FUSION

Blackmagic Fusion Comp Publisher Guide

Guide v1.04 · November 6, 2026 · Blackmagic Fusion Comp Publisher v2.13

Free software under the GNU GPL v3 or later. Not associated with Blackmagic Design. Written Nov 6, 2026, for Blackmagic Fusion Comp Publisher 2.13.

v1.00: first guide. v1.01: updated for 2.09, whose installer shows the same "not associated with Blackmagic Design" notice. v1.02: for 2.11, which is free software under the GNU GPL v3 or later and is on GitHub, with the installer to download. v1.03: for 2.12, which opens on Macs with macOS 12 or later (2.11 needed the newest macOS). v1.04: for 2.13, which sets the comp's frame size from the L1 clip.

Starts every Fusion comp for you. Point it at a folder of shots and it makes one comp per shot, each already holding the shot's L1 clip in a Loader, with the render range and working range set to the clip's length. It is a Mac app that you install once.

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1. Install
Double-click the installer

2. Choose shots
Tick the comps to publish

3. Publish
Open a comp that is ready to work

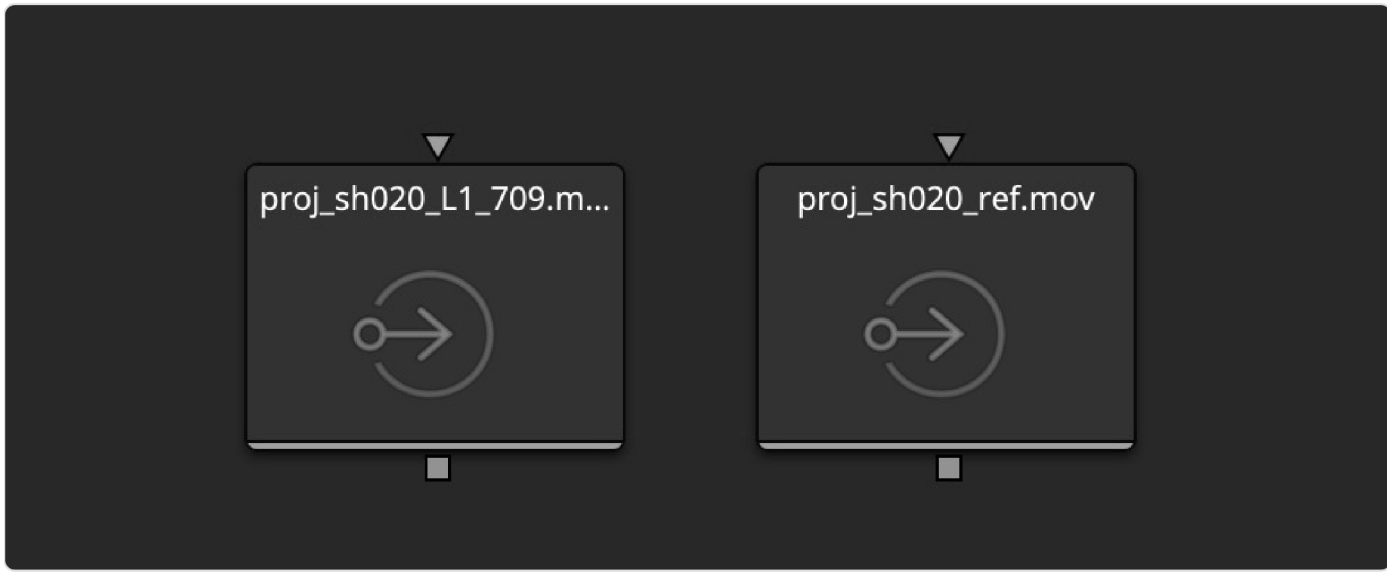
What it does

Say you have a folder of shots, each with its footage in a `source` folder:

```
Shots/
  proj_sh010/
    source/
      proj_sh010_L1.mov
  proj_sh020/
    source/
      proj_sh020_L1_709.mov
      proj_sh020_L1_flat.mov
      proj_sh020_ref.mov
```

After you publish, every shot has a comp named after it, and the footage is already in it:

```
Shots/
  proj_sh010/
    proj_sh010_v001.comp      <- new
    source/ ...
  proj_sh020/
    proj_sh020_v001.comp      <- new
    source/ ...
```



The new comp: the L1 Loader on the left and the reference Loader on the right

Each comp is a copy of your own **template comp**, so it already has your Fusion layout, frame rate and settings. The work that is done for you:

- **The L1 clip is loaded,** through Fusion's `Comp:` path map, so the comp and its footage can move together.
- **The range is set.** The render range and the working range match the clip, so there is no Shift-dragging a Loader to the time ruler.

- **The comp size matches the plate.** The comp's frame size is set to the L1 clip's picture size, so a 2048 x 858 plate does not open in an HD or UHD comp. The frame rate stays your template's.
- **A reference clip is loaded too,** as a second Loader, if the shot has one. It never changes the range.



The range fields after opening a comp: 1001 to 1072, with the working range ending at 1073

What you need

- **An Apple Silicon Mac** (M1 or later) **with macOS 12 or later.** The app does not run on Intel Macs.
- **Blackmagic Fusion** (Fusion Studio) to open the comps. The Publisher does not open Fusion itself.
- **Nothing else.** The app has what it needs inside: there is no Python or ffmpeg to install.

Install

1. **Download** `BlackmagicFusionCompPublisher-<version>.pkg` from the Releases page, then **double-click** it.
2. Follow the installer: press **Continue**, then **Install**, and enter your Mac's password when it asks.
3. Open **Blackmagic Fusion Comp Publisher** from your Applications folder. It sits beside the Blackmagic Fusion folder, so it is easy to find.

To install a newer version, run the new installer: it replaces the old app.

Set up your shots

The Publisher looks inside each shot folder for a folder named `source`. What it finds there decides what goes in the comp.

In `source`	What it is	Loaded as
`proj_sh010_L1.mov`	The L1 clip, a movie	The first Loader
`proj_sh010_L1/proj_sh010_L1.1001.exr`	The L1 clip, an image sequence in its own L1 folder	The first Loader
`proj_sh010_ref.mov`	A reference clip	A second Loader

- **L1 movies:** QuickTime or MP4, named `<shot>_L1.mov`.

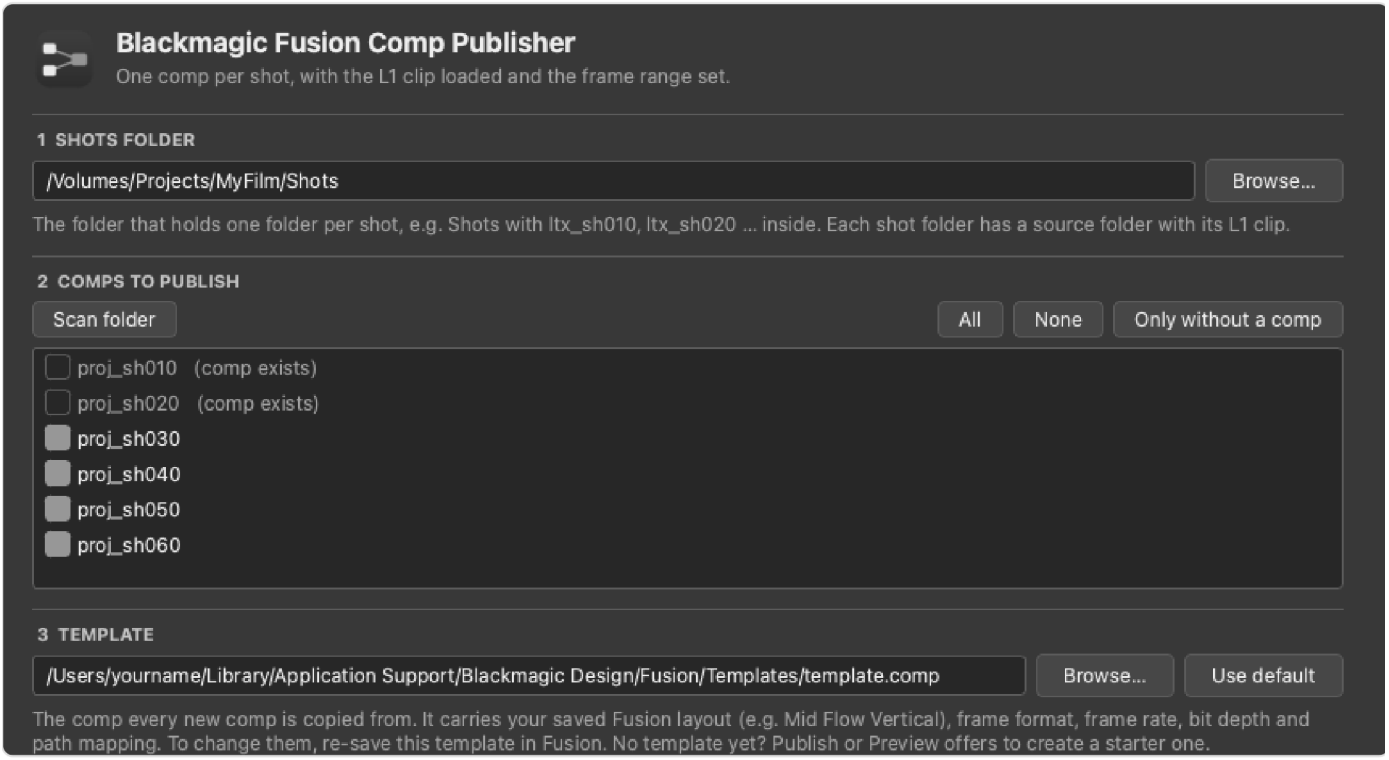
- **L1 image sequences:** EXR, DPX, TIFF, JPEG or PNG. Put the frames in a folder named `<shot>_L1` inside `source`, with names like `proj_sh010_L1.1001.exr`. Only the frames go in that folder. The sequence runs from its first frame number to its last.
- **Colour tags:** when a shot has the same footage in two versions, add a tag to the file name: `proj_sh020_L1_709.mov` and `proj_sh020_L1_flat.mov`. The folder of an image sequence is not tagged, so put the tag in the frame names instead.
- **Reference clips:** a movie, or an image sequence, whose name ends in `_ref`, in lowercase, like `proj_sh020_ref.mov`. It does not have to say L1, because one reference can cover all the layers. The Publisher looks for it **anywhere inside the shot folder**. It expects it at the root of `source`, and that one wins if there is more than one.

Your template

Every new comp is copied from one template comp. It is what carries your Fusion layout (for example Mid Flow Vertical), frame format, frame rate, bit depth and path mapping.

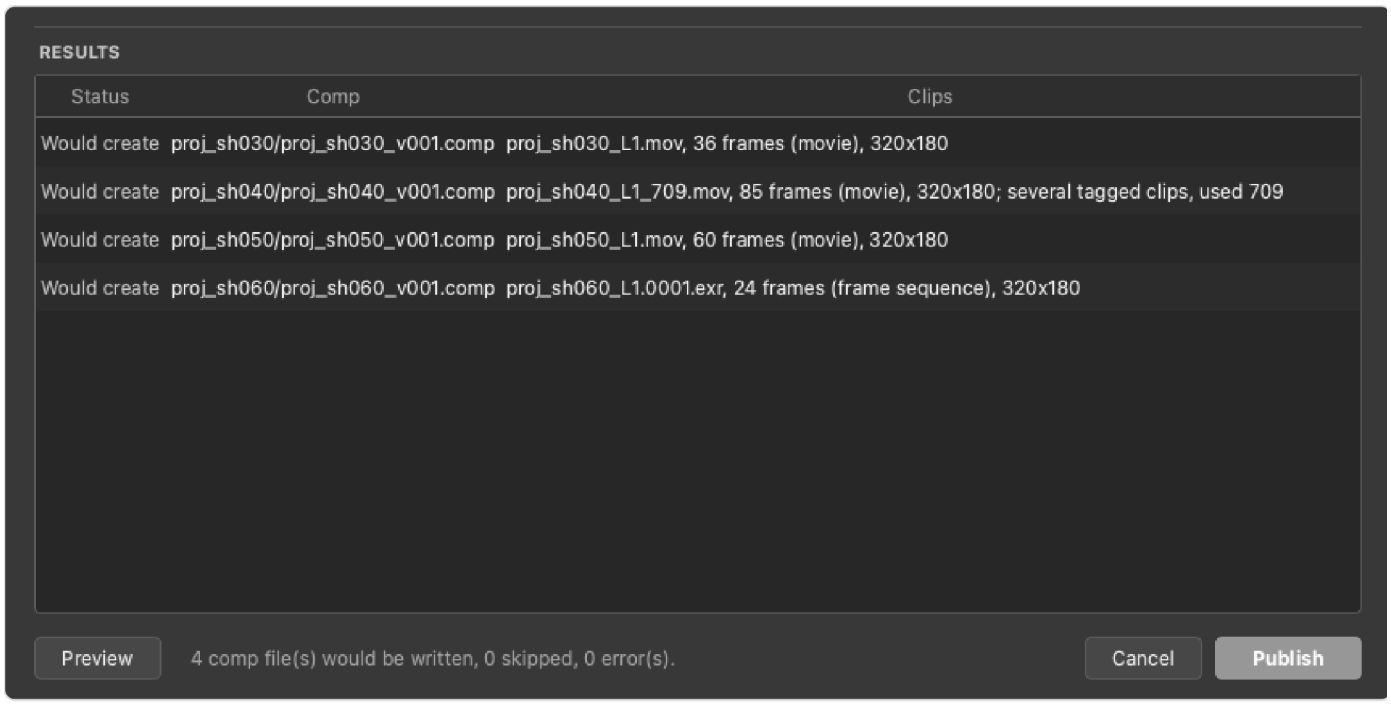
- **Where it lives:** in your user Templates folder, the same place Fusion looks: `~/Library/Application Support/Blackmagic Design/Fusion/Templates/template.comp`. If there is none there, the Publisher uses the system-wide one in `/Library/Application Support/Blackmagic Design/Fusion/Templates`.
- **No template yet?** The first time you press **Preview** or **Publish**, the Publisher offers to create a starter template. It is a plain comp with Fusion's default layout and settings. To make it yours, open it in Fusion, set up your layout and preferences, and save it over the same file.
- **Keep its start frame at 1001.** The comp's first frame comes from the template's render range.

Publish your comps



The top of the window: the shots folder, the comps to publish and the template

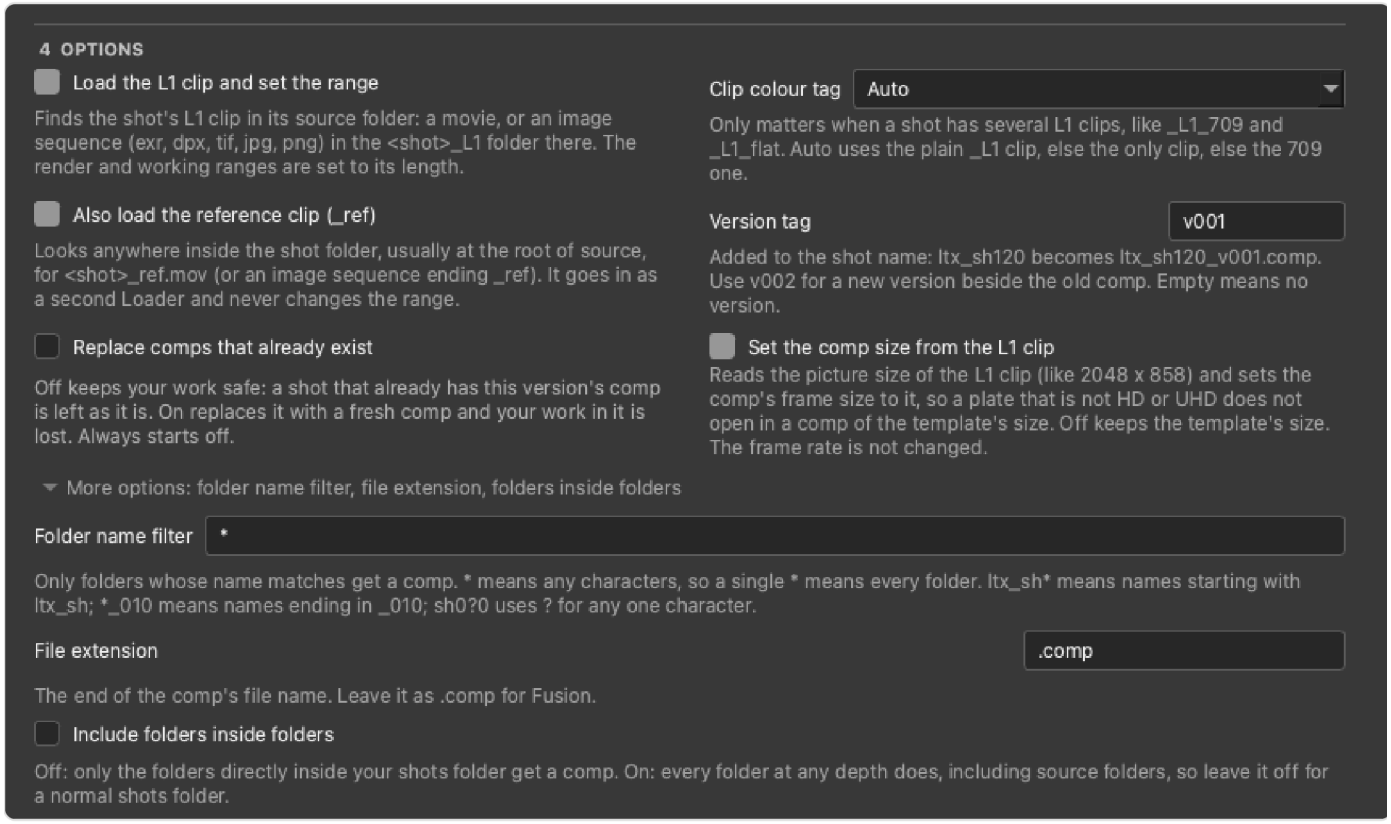
1. **Shots folder.** Press **Browse** and choose the folder that holds your shot folders (the `Shots` folder in the example above).
2. **Comps to publish.** Every shot folder is listed with a tick box. A shot that already has this version's comp starts **unticked**, so a re-run leaves it alone. **All**, **None** and **Only without a comp** change the ticks quickly.
3. **Template.** It is filled in with your default template. Press **Use default** to put it back after you browse to another.
4. **Options.** The next section explains each one. The defaults suit most shots.
5. Press **Preview** to see what would happen without writing a file, then **Publish**.



The results list after a Preview

The **Results** list shows each comp and what went into it: the clip name, how many frames, and notes such as which of two clips was used. Double-click a row to reveal that folder in the Finder. After you publish, **Cancel** becomes **Close**, because there is nothing left to cancel.

What the options do



The options, with More options opened

Option	What it does
Load the L1 clip and set the range	Finds the shot's L1 clip and sets the render and working ranges to its length. Turn it off and each comp is just a copy of your template.
Clip colour tag	Only matters when a shot has several L1 clips, like `_L1_709` and `_L1_flat`. Type or pick the tag to load. **Auto** uses the plain `_L1` clip, else the only clip, else the 709 one. The list shows the tags it found in your shots.
Set the comp size from the L1 clip	Reads the picture size of the L1 clip and sets the comp's frame size to it. Off keeps your template's size. The frame rate is not changed.
Also load the reference clip (_ref)	Adds a second Loader for the shot's `_ref` clip, placed to the right of the L1 Loader.
Version tag	Added to the shot name: `proj_sh020` becomes `proj_sh020_v001.comp`. Use `v002` to start a new version beside the old comp. Empty means no version.
Replace comps that already exist	**Off keeps your work safe:** a shot that already has this version's comp is left as it is. On replaces it with a fresh comp, and the work in the old one is lost. It always starts off.

Option	What it does
Folder name filter (More options)	Only folders whose name matches get a comp. `*` means any characters, so a lone `*` means every folder. `proj_sh*` means names that start with `proj_sh`. `*_010` means names that end in `_010`. `sh0?0` uses `?` for any one character.
File extension (More options)	The end of the comp's file name. Leave it as `.comp` for Fusion.
Include folders inside folders (More options)	Off: only the folders directly inside your shots folder get a comp. On: every folder at any depth does, including `source` folders, so leave it off for a normal shots folder.

The Publisher remembers your choices for next time, except **Replace comps that already exist**.

How the range is set

- **Render range:** the clip's length, starting at 1001. A 72-frame clip gives 1001 to 1072.
- **Working range:** one frame longer, 1001 to 1073. This is what Fusion sets when you Shift-drag a Loader to the time ruler.
- **Always from L1.** A reference clip that is longer or shorter changes nothing. Its Loader starts at 1001 and runs for its own length.

How the size is set

- **Where it comes from:** the L1 clip's own file. For a movie, its picture size. For an image sequence, the first frame's header: the display window of an EXR, or the width and height of a DPX, TIFF, JPEG or PNG. The `_ref` clip never changes it.
- **What changes:** the comp's frame size, so a 2048 x 858 plate gives a 2048 x 858 comp. A plate that is exactly 1920 x 1080 or 3840 x 2160 keeps Fusion's HD 1080 or Ultra HD 2160 name; any other size is named Custom.
- **What does not change:** the frame rate, bit depth and everything else in your template.
- **Pixel shape:** the size is the picture's width and height in pixels. A movie that uses non-square pixels is not adjusted.

Troubleshooting

What you see	What it means
"no L1 clip found; plain template, no Loader"	There is no L1 clip in `source`. The name needs `_L1` in it (`proj_sh010_L1.mov`), and image frames must be in a `<shot>_L1` folder. A name ending in `_ref` is never taken for the L1 clip.

What you see	What it means
"no clip tagged ..., used Auto instead"	The tag you chose is not on that shot, so it used the plain L1 clip, or the only one.
"several tagged clips, used 709"	The shot has `_L1_709` and `_L1_flat` and the tag is on Auto, so it took the 709 one. Choose a tag to override.
"size not read, kept the template's"	The size could not be read from that clip's header, so the comp keeps your template's size. The Loader and the range are still set.
"could not read the frame count of this movie"	The movie file is damaged or an unusual kind. The comp gets your plain template with no Loader.
The comp's Loader shows no picture	The `source` folder was moved apart from the comp. The Loader looks for `Comp:/source/...`, which is the folder the comp is in.
"comp exists" and the shot is unticked	That shot already has this version's comp. Use a new **Version tag**, or turn on **Replace comps that already exist** if you really want a fresh one.
A shot is missing from the list	Check the **Folder name filter**, and press **Scan folder** after changing it.

Where things are saved, and removing it

- **Your options:** `~/.cvt/fusion_comp_publisher.json`. Delete it to start fresh.
- **To remove the app:** drag **Blackmagic Fusion Comp Publisher** from Applications to the Trash. The installer made it, so your Mac asks for your password.
- **Your comps:** the Publisher never deletes or moves any, and it replaces one only when you turn that on.

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Free software

Blackmagic Fusion Comp Publisher is free software under the GNU General Public License, version 3 or later: you may use it, change it and share it. The source code is at github.com/creativevideotips/cvt-fusion-comp-publisher.

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